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Data Revisions Are Not Well Behaved

We document the empirical properties of revisions to major macroeconomic variables in the United States. Our findings suggest that they do not satisfy simple desirable statistical properties. In particular, we find that these revisions do not have a zero mean, which indicates that the initial announcements by statistical agencies are biased. We also find that the revisions are quite large compared to the original variables and they are predictable using the information set at the time of the initial announcement, which means that the initial announcements of statistical agencies are not rational forecasts.

JEL codes: C22, C53, C82

Keywords: forecasting, news and noise, real-time data, NIPA variables.

MOST MACROECONOMIC VARIABLES are substantially revised by statistical agencies in the months after their initial announcements. These revisions generally reflect the arrival of new information that was not available at the time of the initial announcement. Users of data understand the uncertainty surrounding the initial announcement and make their decisions accordingly. If revisions are “well behaved,” by which we loosely mean that they are rational forecast errors, then the arrival of a new revision is not relevant for them. In this paper, however, we will argue that revisions are not, in fact, “well behaved.”

To facilitate the discussion, we will use the following notation. Let y_t^{t+1} denote a statistical agency’s initial announcement of a variable that was realized at time t and y_t^f denote the *final* or *true* value of the same variable. The two objects will be related by the following identity

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$$y_t^f \equiv y_t^{t+1} + r_t^f,$$

where r_t^f is the final revision that is potentially never observed.

From a statistical point of view, we expect the final revision to satisfy three properties in order to consider it well behaved. First, we expect its mean to be zero. This would imply that the initial announcement of the statistical agency is an unbiased estimate of the final value. Second, we expect the variance of the final revision to be small, compared to the variance of the final value. Finally, we expect the final revision to be unpredictable given the information set at the time of the initial announcement. When the final revision is predictable, the initial announcement of the statistical agency is not an optimal forecast of the final value and a better forecast, one with a lower forecast error variance, can be obtained. We summarize these three properties as follows:

$$(P1): \quad E(r_t^f) = 0$$

$$(P2): \quad \text{var}(r_t^f) \text{ is small}$$

$$(P3): \quad E(r_t^f | I_{t+1}) = 0,$$

where I_{t+1} is the information set at the time of the initial announcement. Our goal in this paper is to investigate the validity of these properties for revisions to some major macroeconomic variables in the United States.

We are certainly not the first to analyze the statistical properties of data revisions. Indeed, that macroeconomic data are revised is well understood by economists and various aspects of data revisions have been studied for decades. An important part of the literature considers the question we devote most of this paper to, the predictability of data revisions. Mankiw, Runkle, and Shapiro (1984) assess whether the preliminary announcements of money stock are rational forecasts of the final announcements (news hypothesis) or are observations of the revised series, measured with error (noise hypothesis). A similar analysis was applied to gross national product (GNP) data by Mankiw and Shapiro (1986, henceforth MS). The conclusion from these two studies is that while the revisions to GNP are news, those of money stock data are better characterized as noise. In other words, they find evidence of predictability for the revisions to the money stock data while revisions to GNP data seems to be unpredictable. Mork (1987) and Mork (1990) consider the same question and find predictability in both GNP and money stock revisions using a slightly different methodology.

In a recent paper Faust, Rogers, and Wright (2005) look at the revisions to the gross domestic product (GDP) growth rates for the G-7 countries and find that while for the United States, revisions are only slightly predictable, for Italy, Japan, and United Kingdom, about half the variability of subsequent revisions can be accounted for by information available at the time of the preliminary announcement by using methods similar to Mankiw, Runkle, and Shapiro (1984) and MS.

A recent paper by economists at the Bureau of Economic Analysis (BEA), Fixler and Grimm (2002), analyzes the reliability of National Income Product Account (NIPA) data for the period 1983–2000. It reports mean revisions that are close to those we find in this paper and concludes that they are not significant. As for forecastability, they only consider forecasting a vintage of the data using an earlier vintage, and they conclude that revisions are not predictable. Our methodology as well as our conclusions will be different.

After analyzing some of the basic statistical properties of revisions to a variety of important macroeconomic variables, we find strong evidence against the three properties outlined above. In particular, we find that the unconditional mean of revisions are positive for all variables—significantly so for a majority of them. Moreover, we find that variance of the revisions are quite large compared to the variance of the original data series. We also show that the zero forecast implied by (P3) can be improved significantly in both an *ex post* forecasting exercise and in a real-time forecasting exercise. We find that these results are robust in subsamples, if not stronger since the mid-1980s. Interestingly, we find a larger variability in revisions and a larger degree of predictability in periods that coincide with the decline in volatility that is well-documented for the U.S. economy. We also show that the findings are robust if we group revisions by the quarter of the initial announcement and analyze intermediate revisions.

The rest of this paper is organized as follows. In Section 1, we describe the data used in the paper. In Section 2, we report the unconditional properties of revisions, investigating the validity of (P1) and (P2). In Section 3, we turn to predictability of revisions and consider the validity of (P3). In Section 4, we explore the robustness of our results. We conclude in Section 5. An appendix that provides some details of the analysis and more results is available from the author or on the Internet at www.boraganaruoba.com.

1. DATA

1.1 Data Sources

Most of our data come from the “Real-Time Data Set” (RTDS) produced by the Federal Reserve Bank of Philadelphia.¹ The RTDS records the information set that would be available to someone on the 15th day of the middle month of a quarter starting from the last quarter of 1965 through the last quarter of 2005. It has quarterly observations and quarterly vintages for major NIPA variables such as real and nominal output, consumption, investment, and their subcategories, monetary measures, banking system data, price level, and unemployment rate. It also includes monthly

1. The data set is publicly available on the Internet at <http://www.phil.frb.org/econ/forecast/reaindex.html>. See Croushore and Stark (2001) for the details of the data set. Croushore and Stark (2003) provide some examples of empirical applications using this data set.

observations and monthly vintages on capacity utilization, industrial production, and employment.

Our analysis will focus on eight variables derived from two original NIPA variables (nominal and real output)²—growth of real output, real final sales,³ nominal output and inflation based on output deflator, annual and quarterly—unemployment rate and levels and growth rates of employment, capacity utilization, and industrial production. In Section 4.2, we also summarize our results for revisions to the growth rates of the components of real output in order to understand which components are responsible for the results we report in the paper regarding revisions to real output.

We also put together a small-scale RTDS for this paper using non-farm business labor productivity (measured as output per hour) as announced by the Bureau of Labor Statistics (BLS) in the Monthly Labor Review (MLR) covering 1971–2005.⁴

Overall we have a mixture of 19 monthly and quarterly variables. All of our variables are in percentage terms either by transformation (e.g., growth rates) or by definition (e.g., unemployment rate). All growth rates are expressed in annual terms. More details about the data set and list of variables, their respective samples, observation frequencies, and sources are provided in an appendix available from the author or on the Internet at www.boraganaruoba.com.

1.2 Initial Announcements

Our first task is to derive the initial announcements for each variable, y_t^{t+1} . To that end, we use the first available announcement in the RTDS for date t . In most instances, this corresponds to using the number that appears in the vintage of next quarter, which is the most recent announcement as of the 15th of the middle month of the quarter, 45 days after the end of the quarter.⁵ For variables with monthly vintages, we use the first available announcement after the end of the month, which is typically 15 or 45 days later. For the set of variables we use, it is unlikely that two announcements are made within 45 days following the end of the month, which would cause us to miss the initial announcement.⁶

2. The RTDS uses GNP before 1992 and GDP afterwards, following the “headline variable” announced by the Bureau of Economic Analysis (BEA). As such, we will use the term “output” instead of GNP or GDP.

3. Real final sales is defined as the difference between real output and real change in inventories.

4. Unlike the RTDS, we only recorded the first announcement regarding each quarter and did not attempt to record intermediate revisions.

5. A concern one might have is whether the first number that appears in the RTDS is indeed the first number announced by the statistical agency. For quarterly variables, all of which are announced by the BEA, except for labor productivity, the $t + 1$ vintage captures the “advance” announcement that is indeed the first number announced by the BEA. To be specific, for 2005Q1, for example, the “advance” estimate of the BEA was published on April 28, 2005, the “preliminary” estimate was published on May 26, 2005 and the 2005Q2 vintage of the RTDS would record the information on May 15, 2005, capturing the “advance” estimate and not the “preliminary” estimate. The “flash” estimate which was announced 15 days before the end of the quarter until 1985 is not used in this study.

6. One exception to the otherwise fairly regular announcement schedule of the BEA was during the government shutdown at the end of 1995 where the release of data was delayed. For 1995Q4 initial release we look at the March 1996 issue of Survey of Current Business.

1.3 Defining the Revisions

We define revisions as follows:

$$r_t^h = y_t^{(t+1)+h} - y_t^{t+1},$$

which measures the cumulative revision up to time $t + 1 + h$, which is h periods after the initial announcement.⁷

1.4 Benchmark Revisions

Most of the revisions we observe are due to arrival of new information. However, occasionally (e.g., about every 5 years for NIPA variables) statistical agencies make changes to their methodologies or make statistical changes such as change of base years or seasonal weights. Such revisions are called benchmark revisions. For some variables, such as real output, benchmark revisions are problematic for the users of the data because they would not be able to extract information from these revisions that they can compare with their old information set. To avoid contaminating our analysis with these benchmark revisions, we only focus on growth rates of variables whose levels do not jump up or down following a benchmark revision.

1.5 Defining the Final Revision

In the literature, the final revision is usually defined as the difference between the latest available observation for the variable and its initial announcement.⁸ This may not necessarily be the best choice due the benchmark revisions. It is true that benchmark revisions often use new information (such as Census data that arrive every 10 years) and enhance the existing estimates in addition to all the other methodological changes. However, it is not reasonable to expect a benchmark revision in the 1990s to have some *new* information about 1970s. Moreover, because statistical agencies make changes to the historical data in order to have a consistent variable over time, the benchmark revisions may distort how the economy looks in the past.⁹ This would suggest, therefore, instead of using the latest available revision as the final revision, we should include as many revisions as possible in our final revision in order to include all relevant revisions, but we want to avoid including too many benchmark revisions.

To define the final revision we determine the numbers of periods after which there are no more revisions for each variable, except for benchmark revisions. For some

7. For variables that are growth rates, the revision is defined on the growth rates, rather than computing the growth rate of the revision of the level of the variable.

8. The final revision concept we use in this paper is not related to the “final” announcement of NIPA variables that is announced by the BEA about 3 months after the end of the quarter, following the “advance” and the “preliminary” announcements in the previous months.

9. For example, the weight on goods related to information technology in the 1970s is certainly not the same as that in 1990s. If a benchmark revision applies the same weights to both periods, the picture for the 1970s will be distorted.

variables such as the NIPA variables, the statistical agencies follow a very specific schedule for revisions that makes it very easy for us to define the final revisions. For other variables, we look at the incremental revisions at different horizons and find a pattern in revisions. Essentially, for each variable we find a finite number K , and define the K th revision of the variable as the final revision. For most of the variables we analyze, K roughly corresponds to 3 years. The rest of the paper is devoted to analyzing the statistical properties of these final revisions.

2. UNCONDITIONAL PROPERTIES OF FINAL REVISIONS

In this section we first consider whether revisions to macroeconomic data in the United States satisfy the first two of the three properties we listed in the Introduction. The results are reported in Table 1.

The first column of Table 1 reports the number of observations for each variable. For quarterly variables we have about 37 years of data while for the monthly variables we have between 20 and 40 years of data. The next column reports the mean of the final revision for each variable. We use Newey–West (Newey and West 1987) heteroskedasticity- and autocorrelation-consistent standard errors in computing the test of significance for these means due to the apparent autocorrelated structure of revisions.¹⁰ The results indicate that the means of final revisions for all 19 variables are positive and except for six variables (annual growth of real output and real final sales, quarterly growth of labor productivity, unemployment rate, and two different measures of capacity utilization) they are statistically different from zero. The interpretation of this result is that the initial announcements of the statistical agencies are biased estimates of the final values. In addition to being statistically significant, the means of final revisions are quite large: the numbers range from 0.1% to 1.2%, excluding the unemployment rate. It is worth noting that the average revision for real output growth is between 17 and 26 basis points, depending on the measure, which is economically significant, considering that the average growth rate of real output in this period is about 2.8%. We can conclude that there is strong evidence against ($P1$); i.e., the revisions do not have a zero mean.

The next two columns report the minimum and maximum final revision for each variable. We see that the range of final revisions for all variables are quite large. For example, the final revision of annual real output growth fluctuates between -1.6% and 2.9% while the final revision of annual labor productivity growth fluctuates between -2.9% and 3.3% . The only possible exception is the final revision to unemployment rate and this only fluctuates between -0.2% and 0.2% , which is consistent with the observation that the revisions to the unemployment rate are small and confined to changes in seasonal factors.

10. All statistical tests in this paper uses 10% significance. In some tables we also report the p -values for reference and, where relevant, mark the coefficients with p -values less than 10% with boldface.

TABLE 1
SUMMARY STATISTICS OF FINAL REVISIONS

	<i>N</i>	Mean	Minimum	Maximum	Std. Dev.	Noise / signal	Corr. with initial	A/C (1)
<i>Annual growth variables</i>								
Nominal Output	150	0.31	-1.74	3.61	0.79	0.28	0.09	0.66
Real Output	150	0.17	-1.62	2.94	0.78	0.31	-0.16	0.67
Inflation (Output Deflator)	150	0.12	-0.81	1.12	0.37	0.15	-0.07	0.60
Labor Productivity	134	0.34	-2.85	3.32	1.31	0.79	-0.46	0.65
Real Final Sales	108	0.17	-1.21	1.78	0.70	0.32	-0.23	0.67
Non-Farm Payroll Employment	458	0.13	-0.83	1.22	0.39	0.21	0.36	0.92
Industrial Production (Total Industry)	483	0.41	-2.66	5.40	1.04	0.21	0.05	0.81
Industrial Production (Manufacturing)	336	0.52	-2.70	6.20	1.29	0.23	0.05	0.83
<i>Quarterly growth variables</i>								
Nominal Output	150	0.47	-3.60	7.33	1.71	0.46	-0.02	0.02
Real Output	150	0.26	-3.42	6.56	1.72	0.49	-0.12	-0.04
Inflation (Output Deflator)	150	0.20	-2.56	2.93	0.85	0.33	-0.13	0.02
Labor Productivity	134	0.31	-8.67	6.98	2.99	0.94	-0.40	-0.18
Real Final Sales	108	0.29	-4.09	5.96	1.69	0.52	-0.32	-0.20
<i>Monthly growth variables</i>								
Non-Farm Payroll Employment	458	0.35	-4.85	5.19	1.40	0.52	-0.29	0.12
Industrial Production (Total Industry)	483	1.00	-20.28	24.12	5.17	0.54	-0.13	0.03
Industrial Production (Manufacturing)	336	1.19	-12.81	25.58	5.44	0.55	-0.19	0.06
<i>Variables in percentage</i>								
Civilian Unemployment Rate	150	0.00	-0.20	0.20	0.07	0.05	-0.02	-0.18
Capacity Utilization (Total Industry)	235	0.14	-1.50	2.30	0.81	0.32	-0.23	0.85
Capacity Utilization (Manufacturing)	282	0.11	-2.10	2.40	0.91	0.25	-0.32	0.86

NOTES: All monthly and quarterly growth variables are annualized. Boldface denotes significance at the 10% level. A/C(1) column reports the first order autocorrelation coefficient.

Next, we report the standard deviation of final revisions. because the standard deviation of final revisions by itself may not be very informative of the size of final revisions, we also report the noise-to-signal ratio for final revisions, which is defined as the standard deviation of final revisions divided by the standard deviation of the final value of the variable.¹¹ This statistic, along with the minimum and maximum final revisions, will give us an idea about the size of final revisions relative to the size of the original variables. The numbers we find range from 0.05 to 0.94 with an average of 0.39. Such large numbers suggest that the final revisions are sizable

11. Note that this number is bounded below by zero but not necessarily bounded above by unity due to the possible (negative) correlation between r_t^f and y_t^{f+1} .

compared to the original variables, and we conclude that (*P2*) is not supported by the data.¹²

The next column reports the simple correlation of the final revision with the initial announcement. While it is not possible to talk about a general pattern in terms of sign of the correlations, all but one of the significant correlations are negative. They are as large as -0.46 and the average absolute correlation is 0.19 . This is our first evidence that (*P3*) may not be consistent with the data because the final revisions are correlated with the initial announcements. We take up this issue more rigorously in the next section.

The last column reports the first order autocorrelation coefficients for final revisions. The final revisions to all annual growth variables and both measures of capacity utilization show strong signs of persistence, with positive autocorrelation coefficients between 0.60 and 0.92 . On the other hand, the persistence of the revisions to the quarterly and monthly growth variables is quite weak, and some variables display negative autocorrelation.¹³ We must stress that while the persistence in final revisions suggests the possibility of their predictability, this cannot be used as direct evidence to that effect. The autocorrelated structure documented here cannot be exploited to provide a forecast of r_t^f , because r_{t-1}^f is not realized until $t + K$ and thus is not in the information set of $t + 1$.

To summarize our results from Table 1, we find that the mean final revision is positive for all variables that we consider and statistically significant for most of the variables. We also find that the final revisions are large relative to the original variables. We have some evidence that suggests predictability of revisions. In Section 4, we explore the sources of these results by looking at intermediate revisions, subsamples, revisions to the components of output and analyzing the final revision corresponding to each quarter separately.

3. FORECASTABILITY OF FINAL REVISIONS

Having analyzed the unconditional properties of data revisions in the previous section, we now turn to investigating the validity of (*P3*), which states that the revisions must be unpredictable given the information set at the time of the initial announcement. We start our analysis by revisiting a classic methodology that labels data revisions as “news” or “noise.” Next we conduct two forecasting exercises, an *ex post* exercise that looks at the predictability of final revisions using the full sample and a real-time

12. It is interesting to note that the signal-to-noise ratios for annual growth variables are about half of their counterparts for monthly or quarterly growth variables.

13. One explanation of the persistence in revisions is the particular schedule that revisions follow. As in the case of annual BEA revisions, we often see revisions effecting a number of consecutive periods announced on the same date. If a common information shock such as tax return data or Census data causes the revisions to the variable in these periods, the final revisions will appear correlated.

exercise that attempts to mimic the forecasting problem of a user of statistical data who is trying to forecast final revisions in real time.

3.1 *News versus Noise Revisited*

Two of the most important papers in the literature that analyze the nature of the revisions to macroeconomic variables is MS and Mankiw, Runkle, and Shapiro (1984) where the authors analyze whether the preliminary announcements of GNP and money stock are rational forecasts of the true, or “final” announcements. In this section we replicate some of their analysis with our new (and longer) data set in order to provide a comparison between results from our new data set and the old and well-known results.¹⁴ All results are presented in detail in an appendix available from the author or on the Internet at www.boraganaruoba.com.

In the framework of the aforementioned papers, final revisions can be classified into two categories:

- (i) **Noise:** The initial announcement is an observation of the final series, measured with error. This means that the revision is uncorrelated with the final value but correlated with the data available when the estimate is made.
- (ii) **News:** The initial announcement is an efficient forecast that reflects all available information and subsequent estimates reduce the forecast error, incorporating new information. The revision is correlated with the final value but uncorrelated with the data available when the estimate is made, i.e., unpredictable with using the information set at the time of the initial announcement.

To classify revisions as noise or news, they consider the regressions

$$y_t^{t+1} = \alpha_1 + \beta_1 y_t^f + v_t^1 \quad (1)$$

$$y_t^f = \alpha_2 + \beta_2 y_t^{t+1} + v_t^2, \quad (2)$$

where the joint hypothesis $\alpha_1 = 0, \beta_1 = 1$ would test the noise hypothesis, and the joint hypothesis $\alpha_2 = 0, \beta_2 = 1$ would test the news hypothesis. As can be easily shown (see the appendix available from the author or on the Internet at www.boraganaruoba.com), these hypotheses are mutually exclusive but, they are not collectively exhaustive, that is, we can reject both hypotheses, especially when the unconditional mean of revisions is not equal to zero.¹⁵ When we reject both hypotheses, there is no guidance in the original MS methodology. Using this framework, they conclude that the revisions to GNP (both as level in constant dollars and growth in current dollars) are news and those of money stock data are better characterized as noise, because they reject one and fail to reject the other hypothesis in each case.

14. The methodology of these two papers have been further improved in Mork (1987, 1990) and Kavajecz and Collins (1995). We use the original methodology to be able to compare our results with MS.

15. All these statements are made in the population. Due to sampling errors, we can reject or fail to reject both hypotheses in small samples.

Using the exact subsample that MS have used (1975Q4–1982Q4) we are able to replicate their results, that is we reject the noise hypothesis and fail to reject the news hypothesis for real output growth. However, this conclusion is not robust, even within the same subsample. If the news hypothesis was true, that is if revisions were errors from a rational forecast, then any other explanatory variable that was observed at the time of the initial announcement included in (2) should have a coefficient of zero. When we estimate the following equation

$$y_t^f = \alpha_2 + \beta_2 y_t^{t+1} + \gamma r_{t-1}^1 + v_t^2, \quad (3)$$

where r_{t-1}^1 is the first revision to output of $t - 1$ that is announced at time $t + 1$, we find that γ is statistically significant and, more importantly, the F -test with null hypothesis $\alpha_2 = 0, \beta_2 = 1, \gamma = 0$ is now rejected.¹⁶

Next, we apply the original methodology to all of our variables in the full sample. We find that for all variables except annual growth of real output and real final sales, unemployment and the two measures of capacity utilization, we reject both the news and the noise hypotheses and we are unable to classify revisions as optimal forecast errors or measurement errors for these variables. On the other hand, we reach an equally ambiguous conclusion for annual growth of real output and real final sales and the unemployment rate where we fail to reject both hypotheses. The only two variables for which we have a definite conclusion are the two measures of capacity utilization whose revisions can be classified as noise. When we look at the source of the rejection of both hypotheses, we see that in most of the regressions, the slope coefficient is statistically different from unity and the constant is significantly different from zero in more than half of the regressions. This means that the positive unconditional mean of revisions contribute to this result but it is not the sole source.

To sum up, we find that the original MS results are special because introducing a small variation in the methodology or looking at a longer sample reverses the results.¹⁷

3.2 An Ex Post Forecastability Exercise

In this section we turn to testing if (P3) is supported by the data, that is, if the conditional mean of final revisions with respect to the information set at the time of the initial announcement is zero.¹⁸

16. Although the inclusion of r_{t-1}^1 in the regression seems arbitrary, it must be clear that this is perfectly consistent with the news hypothesis.

17. One factor behind our results may be our increased power in the tests arising from our sample size. Our regressions have at least 150 observations whereas the regressions in MS have 29 quarterly observations.

18. This is essentially the same as testing the news hypothesis since both are a restatement of efficiency (rationality) of preliminary announcements as rational forecast errors must be orthogonal to the information set at the time of the forecast.

To that end, we estimate the following equation.

$$\text{Model 1: } r_t^f = \alpha + \gamma y_t^{t+1} + \sum_{i=1}^s \beta_i r_{t-i}^i + \sum_{i=1}^4 \lambda_i Q_t^i + \delta t + \phi \Delta N_t + \varepsilon_t, \quad (4)$$

where the dependent variable is the final revision and the explanatory variables are a constant, the initial announcement, revisions to past months or quarters announced at time $t + 1$, quarterly dummy variables, Q_t^i , a linear trend and the first difference of the initial announcement of the unemployment rate only for quarterly variables.¹⁹ We use quarterly dummy variables for both quarterly and monthly variables in an effort to limit the number of coefficients estimated. Except for the presence of past revisions as explanatory variables, these equations are very similar to forecasting equations considered in similar studies that analyze the predictability of revisions. We include these revisions to analyze the predictive power of past revisions in explaining future revisions. We also include seasonal dummies in our estimations because there might be some seasonality in the final revisions due to the specific revision schedules of statistical agencies, even though the original series might be de-seasonalized. Finally, we include the quarterly change in the unemployment rate to capture any systematic patterns in revisions due to business cycles. It is important to note that all explanatory variables, including past revisions, are chosen such that they are all known at time $t + 1$ and as such this is a valid forecasting exercise.²⁰

By estimating this equation we are not trying to find the best model for revisions. If that were the case, one would imagine many other variables potentially being relevant, or a multi-variate analysis would be warranted.²¹ Our aim by estimating these equations is to show that we can find a forecasting model that can perform better than the model implied by (P3), one that has a zero conditional mean.

We conduct the exercise using the following algorithm. For each variable, we estimate (4) by considering all possible combinations of explanatory variables.²² Using both Akaike Information Criterion (AIC) and Schwartz Information Criterion (SIC) as a guide we choose the best model for each variable and label this model as Model 1. Using the parameter estimates of this model we get the fitted value of r_t^f , which we denote as $\hat{r}_t^1$.

To understand the marginal contribution of the past revisions to forecasting the final revision, we eliminate them from the model and re-estimate the simple linear regression with the initial announcement, the trend term and seasonal dummy variables. We label this model Model 2 and denote the fitted value as $\hat{r}_t^2$. In Section 2, we showed that for almost all of the variables we consider, the mean final revision is positive

19. Since we do not have intermediate revisions for labor productivity we do not use any revisions as explanatory variables when we estimate the equation for labor productivity.

20. Since a variable that was realized in time $t - k$ is announced for the first time in $t - k + 1$, its k th revision will be in period $t + 1$.

21. For example, one can imagine using information from monthly industrial production to forecast revisions to quarterly output.

22. For most variables, we have 18 explanatory variables with $2^{18} = 262,144$ possible combinations.

and statistically significant. To assess the contribution of all other variables, over and above the gain due to getting the mean right, we define $\hat{r}_t^3$ as the unconditional mean of the final revision. Finally, we consider the forecast of r_t^f based on (P3) and define this case as Model 4 with the forecast given by $\hat{r}_t^4 = 0$ for all t . To summarize we estimate

$$\text{Model 2 : } r_t^f = \alpha + \gamma y_t^{t+1} + \sum_{i=1}^4 \lambda_i Q_t^i + \delta t + \varepsilon_t \quad (5)$$

$$\text{Model 3 : } r_t^f = \alpha + \varepsilon_t$$

$$\text{Model 4 : } r_t^f = \varepsilon_t.$$

Given the forecasts from these four models, we conduct two tests. First, along the lines of our test of rational data revisions, we test for the joint significance of all coefficients in (4). This test will essentially have Model 4 or (P3) as its null hypothesis. We also compare the predictive powers of $\hat{r}_t^1$, $\hat{r}_t^2$, $\hat{r}_t^3$ versus $\hat{r}_t^4$. In order to do so, we compute the root mean squared errors (RMSE) of forecasts from Models 1, 2, and 3 relative to the RMSE of the forecast from Model 4.²³

The results from this exercise are summarized in Table 2. The first panel shows the results when the best model is chosen using AIC and the second panel shows the results when the best model chosen using SIC. While the quantitative results differ slightly as SIC is more conservative in model choice, the qualitative results are the same across the two panels. The second column lists the explanatory variables that are chosen for Model 1 for each variable. Almost all models picked by AIC include at least one past revision that demonstrates the importance of including these variables in predictive regressions. Interestingly the linear trend is important for 10 of the 19 variables we consider. This suggests a potentially time-varying pattern in revisions and we take up this issue in Section 4.1. It is also interesting to note that for the measures of real output growth, the change in unemployment rate is picked as an explanatory variable with coefficients as large as -0.95 (not reported). This means a 1% change in the unemployment rate from $t - 1$ to t would cause bias in the initial announcement of output growth in t as large as 1%, with a downward revision on average during recessions.

The next column reports the p -value of the Wald statistic testing the significance of all coefficients in the regressions. All p -values are less than 5% and in fact most of them are zero, indicating that we can reject the null hypothesis of (P3). In the terminology of the previous section, this means a rejection of the news hypothesis for all of the variables we consider. The next two columns report the R^2 and $\bar{R}^2$ for each regression. In the models chosen by AIC, the R^2 s range from zero (none of the explanatory variables except the constant are relevant) to 0.24 with an average of 0.12 while the average $\bar{R}^2$ is 0.11. For important variables such as annual growth of

23. The relative RMSEs, $RMSE_i/RMSE_4$ are in fact identical to Theil's U -statistic and it is equal to $\sqrt{1 - R^2}$ if the mean of r_t^f is zero and decreases as the latter increases.

TABLE 2
RESULTS OF THE *Ex Post* FORECASTING EXERCISE

Explanatory variables		Wald	R^2	$\bar{R}^2$	RMSE1/ RMSE4	RMSE2/ RMSE4	RMSE3/ RMSE4
Panel A. AIC							
<i>Annual growth variables</i>							
Nominal Output	C, Trend	0.02	0.03	0.03	0.93	0.93	0.94
Real Output	C, Init, Rev9, Rev10, Trend, Unemp	0.01	0.19	0.16	0.89	0.94	0.99
Inflation (Output Deflator)	C, Rev3, Rev6	0.04	0.05	0.04	0.92	0.94	0.95
Labor Productivity	C, Init	0.00	0.21	0.21	0.86	0.86	0.97
Real Final Sales	C, Init, Rev2, Rev6, Trend	0.00	0.24	0.20	0.84	0.89	0.97
Non-Farm Payroll Employment	Init, Trend	0.00	0.12	0.12	0.89	0.89	0.95
Industrial Production (Total Industry)	C	0.00	0.00	0.00	0.93	0.93	0.93
Industrial Production (Manufacturing)	Rev7, Rev10, Trend	0.00	0.06	0.05	0.90	0.93	0.93
<i>Quarterly growth variables</i>							
Nominal Output	C, Rev1, Rev2, Rev5, Trend	0.00	0.09	0.07	0.92	0.95	0.97
Real Output	C, Init, Rev1, Rev3, Rev5, Rev9, Q1, Trend, Unemp	0.00	0.18	0.13	0.90	0.95	0.99
Inflation (Output Deflator)	C, Init, Rev1, Trend	0.00	0.06	0.04	0.94	0.95	0.97
Labor Productivity	C, Init, Q3	0.00	0.18	0.17	0.90	0.91	0.99
Real Final Sales	Init, Rev3, Q3	0.00	0.22	0.21	0.87	0.88	0.98
<i>Monthly growth variables</i>							
Non-Farm Payroll Employment	C, Init, Rev5, Rev6, Q2, Q3, Trend	0.00	0.16	0.15	0.88	0.90	0.97
Industrial Production (Total Industry)	C, Init, Rev1, Rev2, Rev4, Rev7, Rev9, Trend	0.00	0.06	0.04	0.95	0.97	0.98
Industrial Production (Manufacturing)	C, Init, Rev2, Rev4, Rev7, Rev9	0.00	0.11	0.09	0.93	0.96	0.98
<i>Variables in percentage</i>							
Civilian Unemployment Rate	Rev5, Rev8, Rev9	0.00	0.14	0.12	0.93	0.99	1.00
Capacity Utilization (Total Industry)	C, Init, Rev7	0.01	0.11	0.10	0.91	0.95	0.99
Capacity Utilization (Manufacturing)	C, Init, Rev7, Rev10	0.00	0.16	0.15	0.91	0.94	0.99
Panel B. SIC							
<i>Annual growth variables</i>							
Nominal Output	Init	0.00	0.00	0.00	0.94	0.93	0.94
Real Output	C, Init, Trend, Unemp	0.02	0.16	0.15	0.90	0.94	0.99
Inflation (Output Deflator)	C	0.01	0.00	0.00	0.95	0.94	0.95
Labor Productivity	C, Init	0.00	0.21	0.21	0.86	0.86	0.97
Real Final Sales	C, Init, Trend	0.00	0.17	0.15	0.88	0.89	0.97
Non-Farm Payroll Employment	Init	0.00	0.12	0.12	0.89	0.89	0.95
Industrial Production (Total Industry)	C	0.00	0.00	0.00	0.93	0.93	0.93
Industrial Production (Manufacturing)	Rev7, Trend	0.00	0.05	0.04	0.91	0.93	0.93

(Continued)

TABLE 2
CONTINUED

Explanatory variables		Wald	R^2	$\bar{R}^2$	RMSE1/ RMSE4	RMSE2/ RMSE4	RMSE3/ RMSE4
<i>Quarterly growth variables</i>							
Nominal Output	C, Rev5	0.00	0.04	0.03	0.95	0.95	0.97
Real Output	Rev1, Q1	0.01	0.06	0.06	0.96	0.95	0.99
Inflation (Output Deflator)	C	0.01	0.00	0.00	0.97	0.95	0.97
Labor Productivity	C, Init	0.00	0.16	0.15	0.91	0.91	0.99
Real Final Sales	Init, Q3	0.00	0.20	0.19	0.88	0.88	0.98
<i>Monthly growth variables</i>							
Non-Farm Payroll Employment	C, Init, Rev6, Q2, Trend	0.00	0.15	0.14	0.88	0.90	0.97
Industrial Production (Total Industry)	C, Init	0.00	0.02	0.01	0.97	0.97	0.98
Industrial Production (Manufacturing)	C, Init, Rev4, Rev9	0.00	0.08	0.07	0.94	0.96	0.98
<i>Variables in percentage</i>							
Civilian Unemployment Rate	Rev5	0.00	0.08	0.08	0.96	0.99	1.00
Capacity Utilization (Total Industry)	C, Init	0.02	0.09	0.09	0.91	0.95	0.99
Capacity Utilization (Manufacturing)	C, Init, Rev7	0.00	0.15	0.14	0.92	0.94	0.99

NOTES: "C" refers to a constant, "Init" refers to the initial announcement, "RevX" refers to the x th revision, "QX" refers to the dummy variable for the x th quarter and "Unemp" refers to the first difference of quarterly unemployment.

real output, inflation and labor productivity growth, the R^2 s are 0.19, 0.05, and 0.21, respectively. These numbers may not seem too large in other contexts but we think they are economically important in this context.

The last three columns report the RMSEs of Model 1, Model 2, and Model 3 relative to Model 4.²⁴ The average relative RMSE is 0.91, 0.93, and 0.97 for Model 1, Model 2, and Model 3, respectively, while we find numbers as low as 0.84. We also compute that on average our forecasting model provides a 9% improvement over the zero forecast, a 7% improvement over the unconditional mean and using past revisions as explanatory variables provide an improvement of 3%.

To sum up our findings from this *ex post* forecastability exercise, we find that using a very limited information set that is known at time $t + 1$, we are able to predict the final revision that will be realized at $t + K + 1$. Using three different statistics, goodness-of-fit, a Wald test, and relative RMSE, we find that the forecasting model we estimate performs significantly better than a zero forecast that ($P3$) would imply. We conclude that ($P3$) is not supported by the data and that the initial announcements of statistical agencies are not rational forecasts of the true value of variables.

24. All relative RMSEs are less than unity indicating that our forecasting models perform better than a zero forecast for all variables, which is not at all surprising because these are in-sample RMSEs.

3.3 A Real-Time Forecastability Exercise

It is of great interest for practitioners and policy makers to find ways of exploiting the potential forecastability in real time we identified in the previous section.²⁵ In this section, we conduct a real-time forecasting exercise using some of the insights from the previous sections and demonstrate that one can produce a better forecast than a zero forecast in real time.²⁶ The first insight we use is the apparent first-order autocorrelation of final revisions. As we argued above, we cannot use r_{t-1}^K to forecast r_t^K in real-time in the context of a simple linear regression because the former is not realized until K periods later and a K -step-ahead forecast using r_{t-K}^K would not be accurate.²⁷ However, one can usefully exploit this autocorrelation in a state-space framework as will become clear below.²⁸ The second insight we are going to use is the significant negative correlation between the final revision and the initial announcement and that the first difference of the unemployment rate is a variable that was useful in the *ex post* forecasting exercise.²⁹

We proceed as follows. We start in 1984:1, (first quarter or first month, depending on the frequency) and move forward in time using only the information available at each point in time. Using t to denote 1984:1, we want a forecast of the final revision to the variable realized in 1984:1, which we denoted r_t^K , using information as of 1984:2. To that end, we consider the state space defined by

$$\alpha_s = \mu + \rho(\alpha_{s-1} - \mu) + \sigma \nu_s \quad (6)$$

$$Z_s = \alpha_s + \Lambda X_s \quad (7)$$

for $s = 1, 2, \dots, t$ and where α_s is a latent state variable and ν_s is an iid standard normal innovation for the transition equation. X_s is a column of a $2 \times t$ matrix X that contains the initial announcements for y up to and including period t and the first difference of the unemployment rate, and $\Lambda = [\beta_1 \ \beta_2]$ is a coefficient vector. Z_s is an element of a row vector Z that includes all r_s^K , $s \leq t$ observed at the time

25. The most obvious way of doing so would be estimating (4) recursively, using only the available information at each point in time. This would mean dropping the observations in the 3-year period prior to the time of estimation. While in principle there is no problem with doing this, in practice this scheme does not perform well due to parameter instability and sensitivity to the choice of explanatory variables.

26. We must stress that the aim in this section is to simply show that we can find a scheme that works fairly well for the variables we consider. Developing a more general scheme (e.g., using cross-variable relationships) is beyond the scope of this paper.

27. For exposition purposes we treat K as a constant in what follows. In our implementation we use a time-varying K as explained in an appendix available from the author or on the Internet at www.boraganaruoba.com.

28. This general idea has been previously pursued in the literature. Howrey (1978) is one of the first papers to show how one can use the preliminary announcements to get an optimal prediction of the true variable. Conrad and Corrado (1978) apply the Kalman filter for getting better estimates for monthly retail sales. Finally, Tanizaki and Mariano (1995) derive a non-linear and non-Gaussian filter using importance sampling and Monte Carlo integration methods with Kalman Filter and apply this filter to the per capita consumption of the United States.

29. We could use a richer environment with more variables but since we are going to estimate a state space system in a recursive fashion for every period, we want to limit the number of estimated coefficients.

of estimation. This means that the last K elements of the Z vector will have missing values as they are not observed yet. In this linear Gaussian environment with missing values, the Kalman filter is the optimal filter.³⁰ It is useful to point out that if we did not have any explanatory variables in the measurement equation, then the system would collapse to an AR(1) in the final revision and the forecast we are interested would be a K -period-ahead forecast using the last observed final revision. Our method improves over this simpler forecast by adding more recent information through the X matrix.

We estimate the five parameters $(\mu, \rho, \sigma, \beta_1, \beta_2)$ of this state space system via maximum likelihood using standard methods and obtain a filtered estimate of the mean of the state vector $\{a_s\}_{s=1}^t$.

Our object of interest is a forecast of r_t^K , which is the last element of the Z vector. To get this forecast, we use the measurement equation to get

$$\hat{r}_t^5 = a_t + \hat{\beta}_1 y_t^{t+1} + \hat{\beta}_2 \Delta N_t \quad (8)$$

Once the forecast for 1984:1 is obtained, we update t by one period, i.e., add one more row to Z and X , carefully considering the observed information and repeat the procedure above to obtain the forecast for 1984:2.

For investigating the value-added of this relatively complicated forecasting scheme, we also compute the mean of all realized final revisions at time $t + 1$ and denote the forecast using this mean $\hat{r}_t^6$.

In order to assess the forecast accuracy of our real-time model, we use the test developed in Clark and West (2007) that is for nested models. In our context, this test amounts to testing if the time series

$$f_t = (r_t^K)^2 - [(r_t^K - \hat{r}_t^5)^2 - (\hat{r}_t^5)^2] \quad (9)$$

has a mean of zero, which can simply be tested by a regression of f_t on a constant.³¹ The null hypothesis of this test (CW for short) is equal forecast accuracy between the zero forecast and the forecast from our real-time model and the test uses a one-sided alternative.³²

The results from this exercise are reported in Table 3. Columns two through four report the main results where we compare the real-time forecast with the zero forecast. We report the RMSE of the zero forecast $\hat{r}_t^4$ relative to the RMSE from the real-time forecast $\hat{r}_t^5$ and the CW statistic with its p -value. For 10 out of 19 variables, the real-time forecast has a lower RMSE. More importantly, for all but two variables the CW statistic is positive, indicating superior forecast accuracy of the real-time model, with 13 of them statistically significant.³³ For important variables such as annual growth of

30. We use the methods of Durbin and Koopman (2001) for filtering and estimation.

31. We use Newey–West (Newey and West 1987) standard errors with appropriate lags for this test.

32. This test statistic adjusts the more common statistic that involves only the first two terms for the additional number of estimated coefficients in the more complicated model that nests the simpler model.

33. As explained in detail in Clark and West (2007), the CW statistic can be positive even though the RMSE of the more complicated model is higher than the simple model.

TABLE 3
RESULTS OF THE REAL-TIME FORECASTING EXERCISE

	N	RMSE4/ RMSE5	CW statistic	CW p-value	RMSE6/ RMSE5	CW statistic	CW p-value
<i>Annual growth variables</i>							
Nominal Output	76	0.91	0.19	0.07	0.95	-0.04	0.05
Real Output	76	0.95	0.07	0.23	1.01	0.04	0.17
Inflation (Output Deflator)	76	1.04	0.04	0.04	1.01	0.01	0.19
Labor Productivity	72	1.20	0.96	0.00	1.19	0.86	0.00
Real Final Sales	47	0.94	0.04	0.33	0.99	0.00	0.45
Non-Farm Payroll Employment	228	0.80	-0.02	0.24	0.86	-0.03	0.09
Industrial Production (Total Industry)	228	1.06	0.33	0.00	1.00	0.01	0.23
Industrial Production (Manufacturing)	228	1.06	0.40	0.00	0.99	-0.01	0.36
<i>Quarterly growth variables</i>							
Nominal Output	76	0.99	0.33	0.05	1.00	0.02	0.21
Real Output	76	0.98	0.16	0.18	1.01	0.11	0.10
Inflation (Output Deflator)	76	1.04	0.14	0.01	1.02	0.03	0.07
Labor Productivity	72	1.13	2.00	0.00	1.12	1.93	0.00
Real Final Sales	72	1.08	0.72	0.00	1.08	0.54	0.01
<i>Monthly growth variables</i>							
Non-Farm Payroll Employment	228	1.03	0.41	0.00	1.07	0.30	0.00
Industrial Production (Total Industry)	228	1.04	2.91	0.00	1.02	1.04	0.00
Industrial Production (Manufacturing)	228	1.04	4.42	0.00	1.03	1.85	0.00
<i>Variables in percentage</i>							
Civilian Unemployment Rate	76	0.99	0.00	0.13	1.00	0.00	0.11
Capacity Utilization (Total Industry)	156	0.65	0.96	0.00	0.65	1.00	0.00
Capacity Utilization (Manufacturing)	228	0.73	0.15	0.16	0.78	0.26	0.03

NOTES: N denotes the number of periods used for out-of-sample forecasting. RMSE5 refers to the real-time forecast and RMSE4 refers to the zero forecast. Boldface in the RMSE5/RMSE4 column shows entries greater than unity. Boldface in the RMSE4/REMSE5 and RMSE6/RMSE5 columns shows entries greater than unity.

nominal output, inflation and labor productivity, we conclude that using the real-time scheme outlined above would give significant gains in forecasting the final revision over a naive forecast of zero. Looking at the last three columns, we find that for 14 variables the real-time model has additional power over the forecast that only uses the mean, with 11 statistically significant CW statistics.

4. SENSITIVITY ANALYSIS AND FURTHER RESULTS

In this section, we repeat most of the analysis carried out in the previous sections with a number of variations to investigate the source of the findings so far and to see if they are sensitive to these variations. Detailed results are provided in an appendix available from the author or on the Internet at www.boraganaruoba.com.

4.1 Subsamples

As a first sensitivity analysis, we divide the sample into two: before and after 1984, which roughly corresponds to the midpoint of our sample for NIPA variables. It is interesting to do this analysis because we may find that our results are highly dominated by revisions in one of the two subsamples. This may be the case, for

example, as a result of improvements in data collection due to technological progress. However, another equally plausible argument is that technological progress makes data collection harder due to increased variety of goods. This would suggest that as the statistical agencies are struggling to make the necessary corrections, they might create revisions that do not satisfy these three properties. This date is also important because the post-1984 period roughly corresponds to the period where real economic activity in the United States is much less volatile (see, e.g., Stock and Watson 2003). Therefore, it is an independently interesting exercise to analyze the link between this observation and a possible change in the data revision processes.

The results for the unconditional properties of data revisions for the two subsamples are reported in Panel A of Table 4, along with the full sample results for comparison. We find that, out of the 18 variables considered, 11 of them has a higher and statistically significant mean revision in the earlier subsample. Important variables such as real output growth have a much smaller and insignificant mean in the later subsample. However, the mean revision for growth of nominal output or inflation continue to be positive and significant. On the other hand, we find that all variables have higher noise-to-signal ratio in the later subsample, indicating that statistical agencies make bigger revisions.³⁴

The results for the *ex post* forecasting exercise for the two subsamples using AIC are reported in Panel B of Table 4. The first important observation is that for 9 out of 18 variables, the R^2 for both subsamples is greater than the R^2 for the full sample. In general, different sets of explanatory variables are chosen in the two subsamples and in the full sample. Moreover, for all variables the R^2 for at least one subsample is greater than the R^2 for the full sample. We also find that for 11 variables the degree of predictability is bigger in the post-1984 period compared to the pre-1984 period.

To sum up, we find clear evidence of a regime change before and after 1984 as evidenced by the larger noise–signal ratios and the fact that the degree of predictability is higher in subsamples than the full sample.³⁵ The mean revisions are in general lower in the post-1984 period but they continue to be statistically significant for some key variables. We can also safely conclude that the failure of the three properties ($P1$), ($P2$), and ($P3$) we documented in the full sample is not necessarily due to a certain part of the sample. However, we find increased evidence against these properties in the second half of the sample that lends support for the second view about the effect of technological progress on the quality of data described above. The observation that the regime change in data revisions seems to coincide with the “great moderation” is also very interesting.

34. When one looks at the standard deviations of revisions, we see an increase for 6 out of the 18 variables, as much as 18% and an average decline of 13% for all 18 variables. This number is clearly significantly lower than the reduction in volatility of macroeconomic variables documented in the literature. One would expect the magnitude of revisions to fall one-to-one with the magnitude of the true variable. Our results indicate that this is not the case.

35. We do not claim that the break is necessarily in the first quarter of 1984.

TABLE 4

SUMMARY OF RESULTS FOR SUBSAMPLES

Panel A. Unconditional properties

	Full sample			Pre-1984			Post-1984		
	N	Mean	Noise/ signal	N	Mean	Noise/ signal	N	Mean	Noise/ signal
<i>Annual growth variables</i>									
Nominal Output	150	0.31	0.28	74	0.47	0.33	76	0.16	0.36
Real Output	150	0.17	0.31	74	0.33	0.25	76	0.02	0.43
Inflation (Output Deflator)	150	0.12	0.15	74	0.11	0.16	76	0.12	0.36
Labor Productivity	134	0.34	0.79	62	0.36	0.56	72	0.32	1.24
Real Final Sales	108	0.17	0.32	61	0.25	0.26	47	0.06	0.48
Non-Farm Payroll Employment	458	0.13	0.21	230	0.21	0.17	228	0.05	0.26
Industrial Production (Total Industry)	483	0.41	0.21	255	0.39	0.16	228	0.43	0.32
Industrial Production (Manufacturing)	336	0.52	0.23	108	0.49	0.16	228	0.53	0.34
<i>Quarterly growth variables</i>									
Nominal Output	150	0.47	0.46	74	0.68	0.48	76	0.26	0.61
Real Output	150	0.26	0.49	74	0.49	0.42	76	0.03	0.67
Inflation (Output Deflator)	150	0.20	0.33	74	0.19	0.37	76	0.21	0.66
Labor Productivity	134	0.31	0.94	62	0.22	0.88	72	0.39	1.07
Real Final Sales	108	0.29	0.52	61	0.35	0.42	47	0.20	0.81
<i>Monthly growth variables</i>									
Non-Farm Payroll Employment	458	0.35	0.52	230	0.54	0.49	228	0.15	0.59
Industrial Production (Total Industry)	483	1.00	0.54	255	1.14	0.49	228	0.86	0.72
Industrial Production (Manufacturing)	336	1.19	0.55	108	1.71	0.45	228	0.94	0.73
<i>Variables in percentage</i>									
Civilian Unemployment Rate	150	0.00	0.05	74	-0.01	0.05	76	0.01	0.05
Capacity Utilization (Manufacturing)	282	0.11	0.25	54	0.42	0.20	228	0.03	0.32

Panel B. Ex Post Forecasting (AIC)

	Full sample		Pre-1984		Post-1984	
	R ²	RMSE1/ RMSE4	R ²	RMSE1/ RMSE4	R ²	RMSE1/ RMSE4
<i>Annual growth variables</i>						
Nominal Output	0.03	0.93	0.11	0.79	0.17	0.89
Real Output	0.19	0.89	0.15	0.88	0.20	0.89
Inflation (Output Deflator)	0.05	0.92	0.18	0.85	0.25	0.81
Labor Productivity	0.21	0.86	0.13	0.88	0.42	0.75
Real Final Sales	0.24	0.84	0.23	0.79	0.29	0.84
Non-Farm Payroll Employment	0.12	0.89	0.31	0.73	0.09	0.96
Industrial Production (Total Industry)	0.00	0.93	0.05	0.90	0.06	0.91
Industrial Production (Manufacturing)	0.06	0.90	0.47	0.68	0.05	0.91
<i>Quarterly growth variables</i>						
Nominal Output	0.09	0.92	0.16	0.83	0.11	0.93
Real Output	0.18	0.90	0.21	0.87	0.12	0.94
Inflation (Output Deflator)	0.06	0.94	0.09	0.93	0.23	0.84
Labor Productivity	0.18	0.90	0.20	0.87	0.24	0.89
Real Final Sales	0.22	0.87	0.08	0.94	0.36	0.79
<i>Monthly growth variables</i>						
Non-Farm Payroll Employment	0.16	0.88	0.15	0.88	0.20	0.89
Industrial Production (Total Industry)	0.06	0.95	0.06	0.95	0.15	0.89
Industrial Production (Manufacturing)	0.11	0.93	0.28	0.83	0.18	0.89
<i>Variables in percentage</i>						
Civilian Unemployment Rate	0.14	0.93	0.30	0.84	0.20	0.89
Capacity Utilization (Manufacturing)	0.16	0.91	0.78	0.42	0.08	0.96

NOTES: All monthly and quarterly growth variables are annualized. Boldface denote significance at 10% level. There are too few observations for Capacity Utilization (Total Industry) before 1984 to conduct the analysis.

4.2 Summary of Other Results³⁶

In order to understand which revisions, among the many revisions our variables go through, are responsible for the rejection of (*P1*), (*P2*), and (*P3*), we analyze the intermediate revisions, r_t^h for $h < K$, for some key variables where we focus on h values corresponding to the 1-quarter, 1-year and 2-year revisions.³⁷ We find that the mean revision for most of the variables increase with each incremental revision and they are statistically significant. Moreover about half of the volatility of the final revision comes from the revision after one quarter and about 72% of it comes from the 1-year revision. Finally, we reject the news hypothesis for almost all variables. We conclude that most of the intermediate revisions contribute to the rejection of (*P1*), (*P2*), and (*P3*). We can also infer from our results that simply ignoring the initial announcement and using the second or third announcement would not eliminate the problems with revisions.

Next, we analyze the revisions to NIPA variables realized in a certain quarter. We find that for the most part revisions for variables realized in a particular quarter share the same characteristics with the final revision. Moreover, revisions for Q3 variables are more “well behaved” than others and revisions for Q1 variables are the least “well behaved.”

Finally, we repeat our analysis for components of real output in order to identify the source of the results we find for revisions to real output.³⁸ We find that the mean revisions for the annual and quarterly growth of all components are positive, except for three of them. Of these, only three of them are statistically significant but the magnitudes are in general bigger than the mean revision for output. Durables consumption³⁹ and exports stand out as two components with significant (both statistical and economic) mean revisions (0.50 for consumption expenditure on durable goods and 1.33 for exports, compared with 0.17 for output). We also find that all components have larger noise-to-signal ratios as output itself with only two exceptions. Similarly, almost all R^2 s for the estimation of Model 1 are higher and most of the relative RMSEs (Model 1 divided by Model 4) are lower for the components than for output itself. It is interesting to note that the real-time forecastability of the components of output is significantly stronger than output itself, especially between consumption and its subcomponents.

5. CONCLUSION

As users of data, there is nothing we can do about macroeconomic data revisions if they are well behaved. In this paper, we postulate three properties that we expect these

36. Detailed results are available on the Internet at www.boraganaruoba.com.

37. We exclude labor productivity and final sales from this analysis.

38. The components we consider are: consumption (total, durables, non-durables, and services), investment (business fixed and residential), government purchases, exports, and imports.

39. This result is quite significant given the debate concerning measurement of consumer electronics and similar goods whose quality changes quite remarkably in short amounts of time. Our results are at least suggestive that the revisions to components of output that are arguably harder to measure contribute to the results we find in this paper regarding revisions to output.

revisions to satisfy and we find that none of them are satisfied. In particular, we find that the means of final revisions are not zero, indicating that the initial announcements of statistical agencies are biased. We also find that the magnitudes of revisions are quite large compared to the original variables. We further show that the forecast from a forecasting equation is significantly better than a naive zero-forecast, which would be optimal if initial announcements of statistical agencies are optimal forecasts of the final values. This is true for both in an *ex post* exercise and a real-time exercise.

We repeat our analysis for two subsamples and find that while all the findings go through in both samples, the evidence against the three properties seems to be stronger in the second half of the sample. This finding is consistent with the view that technological progress makes collecting data harder due to the difficulty in adjusting the quality of goods in the economy. Another piece of evidence that supports this view is that revisions to durables consumption seem to be an important source of the problem for the results we get regarding the revisions to real output. We also repeat our analysis grouping revisions by the quarter they are first announced and looking at intermediate revisions and find that our results are not driven by one or two sources.

We do not wish to interpret the findings in this paper as failures of the statistical agencies. We believe that these institutions have certain loss functions and use their resources for producing the best possible data and they may be avoiding some other problems at the expense of the problems we outline in this paper. However, whatever the cause of these findings, we think they create problems for the users of the data.

An interesting topic for future research is extending the forecastability analysis to a multivariate framework. There are some interesting and unexpected cross-correlations between revisions to unrelated variables and it would be interesting to explore whether these correlations can be exploited to add to the predictability and forecastability results we obtain in this paper. Moreover, there might be some more expected links between revisions to related variables such as monthly industrial production and quarterly GDP.

Finally, one of the interesting observations from our analysis is the apparent concurrent reduction in the variance of major macroeconomic variables and the increase in the noise-to-signal ratios and predictability. The former is an important observation that has big implications for policy and economic research. Any potential links between these two observations will be of interest to many and is the subject of our ongoing research.

LITERATURE CITED

- Clark, Todd E., and Kenneth D. West. (2007) "Approximately Normal Tests for Equal Predictive Accuracy in Nested Models." *Journal of Econometrics*, 138, 291–311.
- Conrad, William, and Carol Corrado. (1979) "Application of the Kalman Filter to Revisions in Monthly Retail Sales Estimates." *Journal of Economic Dynamics and Control*, 1, 177–98.
- Croushore, Dean, and Tom Stark. (2001) "A Real-Time Data Set for Macroeconomists." *Journal of Econometrics*, 105, 111–30.
- Croushore, Dean, and Tom Stark. (2003) "A Real-Time Data Set for Macroeconomists: Does the Data Vintage Matter?" *Review of Economics and Statistics*, 85, 605–17.

- Durbin, James, and Siem J. Koopman. (2001) *Time Series Analysis by State Space Methods*. Oxford, UK: Oxford University Press.
- Faust, Jon, John H. Rogers, and Jonathan Wright. (2005) "News and Noise in G-7 GDP Announcements." *Journal of Money, Credit and Banking*, 37, 403–20.
- Fixler, Dennis J., and Bruce T. Grimm. (2002) "Reliability of GDP and Related NIPA Estimates." *Survey of Current Business*, 82, 9–27.
- Howrey, E. Philip. (1978) "The Use of Preliminary Data in Economic Forecasting." *Review of Economics and Statistics*, 60, 193–200.
- Kavajecz, Kenneth A. (1994) "The Evolution of the Federal Reserve's Monetary Aggregates: A Timeline." *Federal Reserve Bank of St. Louis Review*, 76:2, 32–66.
- Kavajecz, Kenneth A., and Sean S. Collins. (1995) "Rationality of Preliminary Money Stock Estimates." *Review of Economics and Statistics*, 77, 32–41.
- Mankiw, N. Gregory, David E. Runkle, and Matthew D. Shapiro. (1984) "Are Preliminary Announcements of the Money Stock Rational Forecasts?" *Journal of Monetary Economics*, 14, 15–27.
- Mankiw, N. Gregory, and Matthew D. Shapiro. (1986) "News or Noise: An Analysis of GNP Revisions." *Survey of Current Business*, 66, 20–5.
- Mork, Knut A. (1987) "Ain't Behavin': Forecast Errors and Measurement Errors in Early GNP Estimates." *Journal of Business and Economic Statistics*, 5, 165–75.
- Mork, Knut A. (1990) "Forecastable Money-Growth Revisions: A Closer Look at the Data." *Canadian Journal of Economics*, 23, 593–616.
- Newey, Whitney K., and Kenneth D. West. (1987) "A Simple Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix." *Econometrica*, 55, 703–8.
- Stock, James H., and Mark W. Watson. (2003) "Has the Business Cycle Changed? Evidence and Explanations." *Monetary Policy and Uncertainty*, Federal Reserve Bank of Kansas City, 9–56.
- Tanizaki, Hisashi, and Roberto S. Mariano. (1995) "Prediction of Final Data with Use of Preliminary and/or Revised Data." *Journal of Forecasting*, 14, 351–80.